



# Sprint 1 Planning Document

## Sprint Overview

The main goal of Sprint 1 is to initialize the Next.js project architecture, establish the database connection securely, and implement the authentication lifecycle (Registration, Login, Profile Management), with the core application layout. By the end of this sprint, our team will have a functional skeleton of the application.

**Scrum Master:** Logan Wilkinson

**Meeting Schedule:** Tuesdays & Thursdays @ 4:30 PM (Discord)

## Risks and Challenges

With everyone focusing on their own user story simultaneously, we risk merge conflicts in global config files like layout.tsx. Plus, some of us are new to the App Router, so getting the API routes right could take a second to figure out. We also have to make sure that all members can connect locally to the shared MongoDB Atlas cluster without any IP whitelisting issues.

## 2. Current Sprint Detail

### User Story #1: User Registration

"As a user, I would like to be able to create an account."

- **Owner:** Xavion Marable
- **Estimated Hours:** 6 Hours
- **Tasks:**
  - Create the **User** schema in Mongoose (email, password, etc.) (2 hrs)
  - Build the **POST /api/auth/register** route and the Sign-Up Form UI with Zod validation. (3 hrs)
  - Write unit tests to make sure any duplicate emails get blocked and valid accounts are created. (1 hr)
- **Acceptance Criteria:**
  - **Given** that I'm on the sign-up page, **When** I enter a valid email/password, **Then** a new user document is created.
  - **Given** that I try to sign up with an email that's already taken, **When** I hit submit, **Then** show a "User already exists" error.
  - **Given** that I type in a password shorter than 6 characters, **When** I try to submit, **Then** the form shouldn't let me proceed.

### User Story #2: Profile Management

"As a user, I would like to be able to edit my profile details, like name, school, location, and profile picture."

- **Owner:** Xavion Marable

- **Estimated Hours:** 9 Hours
- **Tasks:**
  - Set up Multer/Cloudinary middleware for image uploads. (2 hrs)
  - Build the **PATCH** `/api/user/me/profile` route to handle text and image form data. (3 hrs)
  - Create the “Edit Profile” UI with an image previewer and loading state. (2 hrs)
  - Write unit tests to make sure any duplicate emails get blocked, profile changes are saved to MongoDB, and to make sure non-image files are rejected and text updates continue working. (2 hrs)
- **Acceptance Criteria:**
  - **Given** I am logged in, **When** I edit my school to “Purdue”, **Then** the change appears on my profile immediately.
  - **Given** I upload a file bigger than 5MB or a non-image file, **When** I try to save, **Then** show a “File too large/Invalid type” error.
  - **Given** I change my profile picture, **When** the upload finishes, **Then** the new image replaces the old one in the navbar and profile page.

### User Story #3: User Account Login and Management

“As a user, I would like to be able to login and manage my account settings.”

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 13.5 Hours
- **Tasks:**
  - Build the **PATCH** `/api/user/me` (2 hrs)

- Build the **POST** `/api/auth/login`. (3 hrs)
- Make the Login Form UI. (3 hrs)
- Make the Account Settings UI. (3hrs)
- Write a unit test to make sure any incorrect logins get rejected and account settings are updated accordingly. (1.5 hrs)
- **Acceptance Criteria:**
  - **Given** that I'm on the sign-in page, **When** I enter a valid email/password, **Then** the user is logged in and sent to the front page.
  - **Given** that I'm on the account settings page, **When** I change any settings, **Then** the change will take effect and be reflected on the account immediately.
  - **Given** that I try to sign in with incorrect info, **When** I hit submit, **Then** show a "Incorrect Email or Password" error.
  - **Given** I am logged in, **When** I edit my account settings, **Then** the change appears on my account immediately.
  - **Given** that I type in an incorrect password, **When** I try to submit, **Then** the form shouldn't let me proceed.

## User Story #4: Password Reset

**"As a user, I would like to be able to reset my password via email/sms if I forget it."**

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 9.5 hours
- **Tasks:**
  - Build the **POST** `/api/auth/password-reset` (2 hrs)

- Build the **POST**  
`/api/auth/password-reset/confirm`. (3 hrs)
- Make the password reset Form UI with ZOD validation. (3 hrs)
- Write a unit test to make sure any incorrect emails for password reset get rejected. (1.5 hrs)
- **Acceptance Criteria:**
  - **Given** that I'm on the forgot my password page, **When** I enter a valid email/sms, **Then** a password reset link is sent to the selected method of communication.
  - **Given** that I try to input an email or phone number that is not attached to a user account, **When** I hit submit, **Then** show a "User does not exist" error.
  - **Given** I am visiting the password reset link, **When** I input a new password, **Then** the change takes effect immediately and allows login with the new password.
  - **Given** that I type in a password shorter than 6 characters or that doesn't match, **When** I try to submit, **Then** the form shouldn't let me proceed.
  - **Given** that I try to log in with my old password, **When** I try to submit, **Then** the password should not work.

**User Story #5: As a user, I would like to be able to input my travel preferences.**

- **Owner:** Shreyansh Tehanguria
- **Tasks:** 10 hrs
  - Create Trip/TripSettings schema in Mongoose (origin, destination, dates, budget, transport, must/avoid, interests, createdBy, groupId) **(2 hrs)**

- Build **POST** `/api/trips/create` (or `/api/trip-settings`) to save trip settings + basic validation (**2 hrs**)
- Build **GET** `/api/trips/me` (or `/api/trips/{id}`) to fetch saved trip settings for editing/reuse (**1.5 hrs**)
- Create Trip Settings UI page + form (Zod validation, dropdowns, multi-city toggle, etc.) (**3 hrs**)
- Connect UI → API (save, load existing trip settings, show success/error states) (**1 hr**)
- Write unit tests for:
  - creates trip settings successfully
  - rejects invalid payload (missing origin/destination/dates, negative budget, etc.)
  - ensures user can only access their own trips / group trips (**0.5 hrs**)
- **Acceptance Criteria:**
  - **Given** I'm on the Trip Settings page, **When** I submit valid settings, **Then** a TripSettings document is created and saved.
  - **Given** I submit missing/invalid fields (ex: no destination or negative budget), **Then** the form blocks submission and shows an error.
  - **Given** I previously saved trip settings, **When** I reopen the page, **Then** my saved settings load correctly.
  - **Given** I try to access someone else's trip settings, **then** the API denies access (401/403).

## User Story #6: Group Messaging

**“As a user, I would like to be able to message the members of my travel group.”**

- **Owner:** Logan Wilkinson

- **Estimated Hours:** 10.5 hours
- **Tasks:**
  - Build **POST** `/api/groups/{id}/messages` (3 hrs)
  - Build **GET** `/api/groups/{id}/messages`. (2 hrs)
  - Create **Message** schema in Mongoose. (1 hrs)
  - Make the messaging UI. (3 hrs)
  - Write a unit test to make sure messages are sent and fetched correctly. (1.5 hrs)
- **Acceptance Criteria:**
  - **Given** that I'm on the messaging UI, **When** I enter a message and submit, **Then**, the message will be pushed and updated locally.
  - **Given** that I'm on the messaging UI, **When** I enter a message and submit, **Then**, the message will be pushed and updated on other group members clients.
  - **Given** that I visit the messaging UI, **When** it is opened, **Then** 50 messages between the current group are populated.
  - **Given** that I visit the messaging UI, **When** I scroll up, **Then** 50 more messages between the current group are populated.
  - **Given** that I try to input an empty message, **When** I hit submit, **Then** show a "Message cannot be empty" error.

## User Story #7:

**"As a user, I would like to be able to access a shared calendar for my travel group."**

- **Owner:** Emilio Guzman
- **Tasks:** 15 hours

- Create **CalendarEvent** schema in Mongoose (title, description, startTime, endTime, location, eventType, createdBy, groupId, source = manual/itinerary, externalId, timezone, updatedAt) **(2 hrs)**
- Build **POST**  
`/api/groups/:groupId/calendar/events` to create a calendar event with validation (time range, required fields, group membership) **(2 hrs)**
- Build **GET**  
`/api/groups/:groupId/calendar/events?from=&to=` to fetch events for a date range (supports weekly/monthly view) **(1.5 hrs)**
- Build **PUT**  
`/api/groups/:groupId/calendar/events/:eventId` to edit an event (only creator or group admin can edit) **(2 hrs)**
- Build **DELETE**  
`/api/groups/:groupId/calendar/events/:eventId` to remove an event (permission check) **(1 hr)**
- Integrate Places search/autocomplete for adding locations (search bar, select result, auto-fill address/coords) **(2 hrs)**
- Create Must-Haves UI page (list + filters, add modal, details view, priority selector, status voting/approval UI) **(3 hrs)**
- Connect UI → API (add/remove/update, optimistic updates, show errors and duplicate warnings) **(1.5 hrs)**

- **Acceptance Criteria:**



- **Given** I am a member of a travel group, **When** I open the Shared Calendar page, **Then** I see all group events in the selected date range.
- **Given** I am on the Shared Calendar page, **When** I create a valid event (title + time range), **Then** the event is saved and appears for all group members.
- **Given** I submit an event with invalid times (end before start) or missing required fields, **When** I click save, **Then** the form blocks submission and shows an error.
- **Given** I am not a member of the group, **When** I request the group calendar events endpoint, **Then** the API denies access (401/403).
- **Given** an event exists, **When** I edit or delete it as the creator (or admin), **Then** the change is persisted and visible to all members.

## User Story #8:

**“As a user, I would like to be able to add activities/locations as must-haves.”**

- **Owner:** Emilio Guzman
- **Tasks:**
  - Create **MustHave** schema in Mongoose (groupId, tripId, placeId, name, category, address, lat, lng, notes, priority, addedBy, status = proposed/approved/rejected, createdAt) **(2 hrs)**
  - Build **POST** `/api/groups/:groupId/must-haves` to add a must-have (validate group membership + deduplicate by **placeId** or name+address) **(2 hrs)**

- Build **GET** `/api/groups/:groupId/must-haves` to fetch list + filters (status, category, priority) **(1.5 hrs)**
- Build **PUT** `/api/groups/:groupId/must-haves/:id` to update notes/priority/status (creator/admin permission rules) **(2 hrs)**
- Build **DELETE** `/api/groups/:groupId/must-haves/:id` to remove a must-have item (creator/admin only) **(1 hr)**
- Integrate Places search/autocomplete for adding locations (search bar, select result, auto-fill address/coords) **(2 hrs)**
- Create Must-Haves UI page (list + filters, add modal, details view, priority selector, status voting/approval UI) **(3 hrs)**
- Connect UI → API (add/remove/update, optimistic updates, show errors and duplicate warnings) **(1.5 hrs)**
- Add “use must-haves in itinerary generation” integration point (ensure must-haves are included in the prompt/input payload to itinerary generator) **(1.5 hrs)**
- **Acceptance Criteria:**
  - **Given** I am a member of a group, **When** I add a location/activity as a must-have, **Then** it is saved and appears in the group must-have list.
  - **Given** I try to add the same must-have twice (same placeId or same name+address), **When** I submit it, **Then** the system prevents duplicates and shows a warning.

- **Given** I update the priority/notes/status of a must-have, **When** I save changes, **Then** the updated values are persisted and visible to the group.
- **Given** I am not part of the group, **When** I call the must-haves API endpoints, **Then** access is denied (401/403).
- **Given** there are must-haves saved for the trip, **When** the itinerary is generated, **Then** the generated plan includes (or strongly prioritizes) the must-have items.

## User Story #9: Create Travel Groups

"As a user, I would like to be able to create travel groups."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 8 Hours
- **Tasks:**
  - Create the **TravelGroup** schema in Mongoose (groupId, groupName, leaderID, membersList, ledger, chatLogs). **(2 hrs)**
  - Build the **POST** **/api/groups/create** route: validate auth, set creator as leader, add creator to membersList, save to MongoDB. **(2 hrs)**
  - Create the "Create Group" UI (group name, optional description) with Zod validation. **(2 hrs)**
  - Wire the form to the API and show success (redirect to group page) or error ("Name required" / server error). **(1 hr)**
  - Write unit tests: group is created with leader in membersList, duplicate group name allowed but unique groupId, unauthenticated requests rejected. **(1 hr)**

- **Acceptance Criteria:**

- Given that I'm logged in and on the Create Group page, When I enter a valid group name and submit, Then a new travel group document is created and I am the leader and a member.
- Given that I submit without a group name, When I hit submit, Then the form shows a "Group name is required" error and does not submit.
- Given that I'm not logged in, When I try to create a group, Then the request is rejected (401) and I am prompted to log in.

## **User Story #9: Manage My Travel Group**

**"As a user, I would like to be able to manage my travel group."**

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 10 Hours
- **Tasks:**

- Build **GET** `/api/groups` (or `GET /api/user/me/groups`) to return all groups the current user is a member of. (1.5 hrs)
- Build **GET** `/api/groups/{id}` to return a single group's details (name, members, leader); ensure requester is a member (403 otherwise). **(1.5 hrs)**
- Build **PATCH** `/api/groups/{id}` to update group name (and optional fields); restrict to group leader. **(2 hrs)**
- Build **POST** `/api/groups/{id}/members` to add a member by email or userID; restrict to leader, validate user exists and is not already in group. **(2 hrs)**

- Create the "Manage Group" / group settings UI: view members, edit group name, invite/add members, only show edit controls if the current user is leader. **(2.5 hrs)**
- Write unit tests: only leader can update group and add members; members can view; non-members get 403. **(1 hr)**
- **Acceptance Criteria:**
  - Given that I am the group leader, When I change the group name and save, Then the group name is updated and reflected for all members.
  - Given that I am the group leader, When I add a member by email, Then that user is added to membersList and can see the group.
  - Given that I am a member but not the leader, When I open the group page, Then I can see the group details and members but not edit name or add/remove members.
  - Given that I am not a member of the group, When I request **GET /api/groups/{id}**, Then I receive 403 Forbidden.

## User Story #10: Leave My Travel Group

**"As a user, I would like to be able to leave my travel group."**

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 5 Hours
- **Tasks:**
  - Build **DELETE /api/groups/{id}/members/me** (or **POST /api/groups/{id}/leave**) to remove the authenticated user from membersList. **(2 hrs)**

- If the user is the leader, either transfer leadership to another member or prevent leave until leadership is transferred (per design: "transfer leadership" exists). Define rule: e.g. leader must transfer first, or auto-assign to next member. **(1 hr)**
- Add a "Leave group" button on the group/settings UI; show confirmation dialog ("Are you sure? You will lose access to this group."). **(1 hr)**
- Write unit tests: member is removed from membersList after leave; leader cannot leave until leadership is transferred (or document auto-transfer behavior). **(1 hr)**
- **Acceptance Criteria:**
  - Given that I am a group member and not the leader, When I click "Leave group" and confirm, Then I am removed from the group and no longer see it in my groups list.
  - Given that I am the group leader, When I try to leave, Then either I am prompted to transfer leadership first, or leadership is automatically transferred to another member and then I leave (choose one and document in the story).
  - Given that I have left the group, When I try to access the group page or group API, Then I receive 403 Forbidden.

## **User Story #11: Friend Connections**

**"As a user, I would like to be able to connect with new people by finding and adding them as friends."**

- **Owner:** Xavion Marable
- **Estimated Hours:** 10 Hours

- **Tasks:**

- Update **User** Schema to include **friends** array and create **FriendRequest** schema. (2 hrs)
- Build **POST** `/api/friends/request` and **PATCH** `/api/friends/accept` API routes. (3 hrs)
- Build the “Find Friends” UI with a search bar that queries users by email. (3 hrs)
- Write unit tests to make sure users can’t add themselves or send duplicate requests. (2 hrs)

- **Acceptance Criteria:**

- **Given** I search a user by email, **When** I click “Add”, **Then** a pending request is created.
- **Given** I have a pending request, **When** I accept it, **Then** we both appear in each other’s friends lists.
- **Given** I try to add someone I am already friends with, **When** I click “Add”, **Then** the system blocks the request.

## User Story #12: Student Verification

**"As a user, I would like to be able to verify my student status (.edu email) to unlock student filters."**

- **Owner:** Xavion Marable

- **Estimated Hours:** 5 Hours

- **Tasks:**

- Update **User** Schema to include **isStudentVerified** boolean and **.edu** email validation logic. (1 hr)
- Build **POST** `/api/auth/verify-student` to send a verification code to the edu email. (2 hrs)

- Create the “Verify Student Status” UI in the settings page. (2 hrs)
- **Acceptance Criteria:**
  - **Given** I enter a valid purdue.edu email, **When** I click verify, **Then** I receive a 6-digit code.
  - **Given** I enter the correct code, **When** I submit, **Then** my profile shows a “Verified Student” badge.
  - **Given** I try to use a gmail.com address, **When** I submit, **Then** the system rejects it.

## User Story #13: Itinerary Interaction/Modification

**"As a user, I would like to be able to see reviews for places people have travelled.**

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 11 Hours
- **Tasks:**
  - Extend Activity schema to store review data (rating, reviewCount, individual reviews or cached reviews) (2 hrs)
  - Build `GET /api/activities/{activityId}/reviews` to fetch reviews for a place (2 hrs)
  - Integrate external reviews source (Places API / stored dataset) and cache results (3 hrs)
  - Create Reviews UI section (scrollable list, rating display, loading states) (2 hrs)
  - Handle errors and empty review cases gracefully (1 hr)
  - Unit tests for valid access, invalid activityId, and authorization (1 hr)
- **Acceptance Criteria:**



- Given I open an activity, when reviews load, then I can see a list of reviews for that place.
- Given a place has no reviews, then a friendly message is displayed.
- Given an invalid activityId, then the API returns an error and no data is shown.

## User Story #14: Itinerary Interaction/Modification

**"As a user, I would like to be able to see a summary of reviews for activities."**

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 10 Hours
- **Tasks:**
  - Add summary fields to Activity (average rating, sentiment summary, highlights) (2 hrs)
  - Implement review aggregation/summary generator (LLM or computed statistics) (3 hrs)
  - Build `GET /api/activities/{activityId}/review-summary` endpoint (2 hrs)
  - Update UI to display summary card (rating, pros/cons, quick insights) (2 hrs)
  - Cache generated summaries to reduce repeated computation (1 hr)
  - Unit tests for summary generation and fallback cases (1 hr)
- **Acceptance Criteria:**
  - Given I view an activity, then I see a summarized overview of reviews.

- Given review data exists, then the summary reflects the overall sentiment.
- Given review data cannot be generated, then a fallback message is shown.

### 3. Backlog (Future Sprints)

#### User Account

- ~~As a user, I would like to be able to create an account.~~
- ~~As a user, I would like to be able to login and manage my account settings.~~
- ~~As a user, I would like to be able to edit my profile details, like name, school, location, and profile picture.~~
- ~~As a user, I would like to be able to reset my password via email/sms if I forget it.~~
- As a user, I would like to be able to link my external calendar (Google/Outlook) to sync trip dates.
- ~~As a user, I would like to be able to verify my student status (.edu email) to unlock student filters.~~
- As a user, I would like to be able to manage my notification preferences (Email, SMS, Push) for trip updates.

#### Travel Groups

- ~~As a user, I would like to be able to create travel groups.~~
- ~~As a user, I would like to be able to manage my travel group.~~
- ~~As a user, I would like to be able to leave my travel group.~~
- ~~As a user, I would like to be able to access a shared calendar for my travel group.~~

- As a group leader, I would like to be able to transfer my leader's permission to another member.
- As a group leader, I would like to be able to remove specific people from the travel group.
- ~~As a user, I would like to be able to message the members of my travel group.~~

## **Trip Creation**

- ~~As a user, I would like to be able to input my travel preferences.~~
- ~~As a user, I would like to create a budget for the trip.~~
- ~~As a user, I would like to be able to specify transportation preferences.~~
- As a user, I would like to be able to add activities/locations as must haves.
- As a user, I would like to be able to add activities/locations to avoid.
- As a user, I would like to be able to generate a random location based on budget.
- As a user, I would like to be able to see recommendations/extra info for activities.

## **Itinerary Interaction/Modification**

- ~~As a user, I would like to be able to see reviews for places people have travelled.~~
- ~~As a user, I would like to be able to see a summary of reviews for activities.~~
- As a user, I would like to be able to visit sites for more information about activities.

- As a user, I would like to be able to go to activity booking sites via external links.
- As a user, I would like to be able to find temporary bag storage.
- As a user, I would like to be able to have an offline itinerary view.
- As a user, I would like to be able to see a visual map representation of the itinerary.
- As a user, I would like to be able to have an offline map view.
- As a user, I would like to be able to create reminders for my travels.
- As a user, I would like to be able to access an SOS button that shows emergency numbers.
- As a user, I would like to be able to write reviews for places I've visited.
- As a user I would like to be able to create "Polls" within the group to vote on destination options.
- As a user, I would like to be able to assign roles (Admin, Viewer) to group members.
- As a user, I would like to be able to view a "Group Board" for pinned announcements.
- As a user, I would like to be able to generate alternative plans.
- As a user, I would like to be able to make changes to only one part of an itinerary.
- As a user, I would like to be able to click a button to regenerate specific parts of the itinerary.
- As a user, I would like to be able to click a button to remove options.

- As a user, I would like to be able to vote on specific activity options created by the itinerary generator.
- As a user, I would like to be able to retrieve previous itineraries.
- As a user, I would like to be able to post itineraries visible to all app users.
- As a user, I would like to be able to share itineraries via link.
- As a user, I would like to be able to export itineraries to my preferred calendar.
- As a user, I would like to be able to select specific “Vibe” tags (Nightlife, Nature, Historical) to guide the itinerary generator.
- As a user, I would like to be able to lock specific activities I like so they don’t change during regeneration.
- As a user, I would like to be able to drag and drop activities to reorder the schedule visually.
- As a user, I would like to generate a “Rainy Day” alternative for my plans.
- As a user, I would like to be able to filter generated locations based on any accessibility requirements.
- As a user, I would like to be able to view a 10-day weather forecast in the itinerary.

## **Itinerary Budget**

- As a user, I would like to be able to post a shared cost to the group ledger.
- As a user, I would like to be able to split the costs with my group.
- As a user, I would like to be able to view a list of what everyone owes at the end of the trip.

- As a user, I would like to be able to request payment from specific group members for a shared cost.
- As a user, I would like to be able to send a payment confirmation notice to a member after paying them back.

## **Communication**

- ~~● As a user, I would like to be able to connect with new people by finding and adding them as friends.~~
- As a user, I would like to be able to create a shared photo album for trips.
- As a user, I would like to be able to send text notifications to my group.
- As a user, I would like to be able to send email notifications to my group.
- As a user, I would like to be able to send in-app notifications to my group.
- As a user, I would like to be able to receive text notifications about my travel.
- As a user, I would like to be able to receive email notifications about my travel.
- As a user, I would like to be able to receive in-app notifications about my travel.
- As a user, I would like to be able to share my real-time location with added friends.